

Fundamentals of Electrical and Electronics Engineering (FEEE)

SUB CODE: ECE1002

T-P-C: 3-2-4

Lecture-7

Faculty

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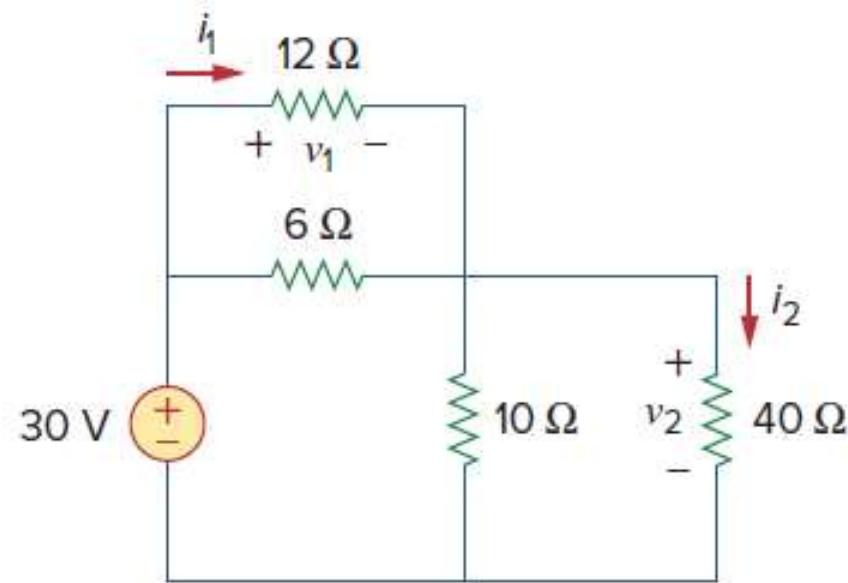
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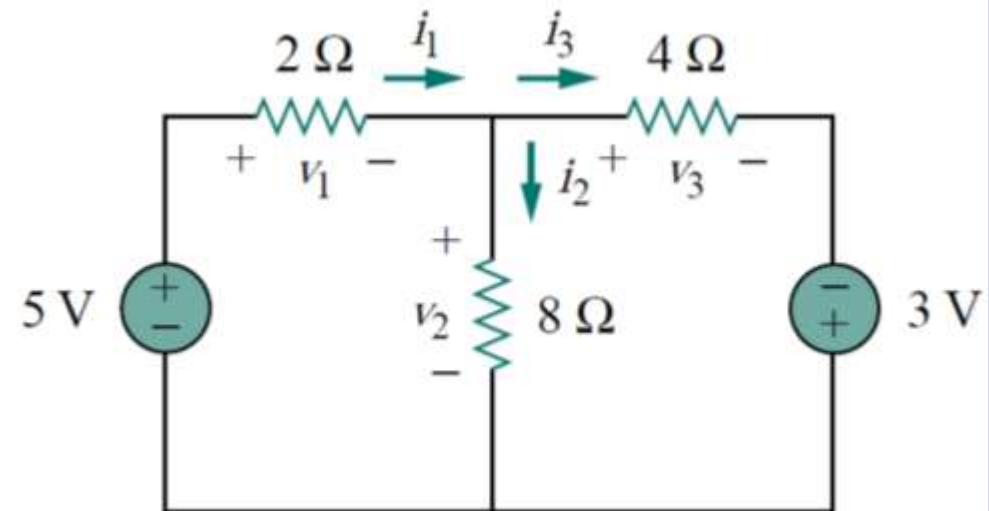
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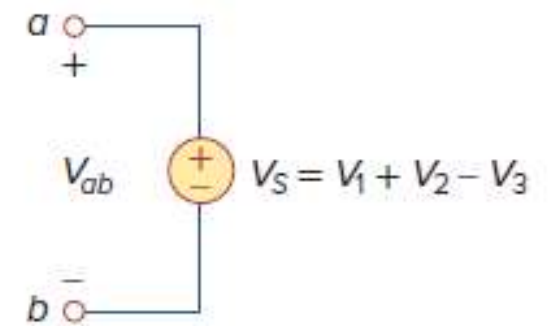
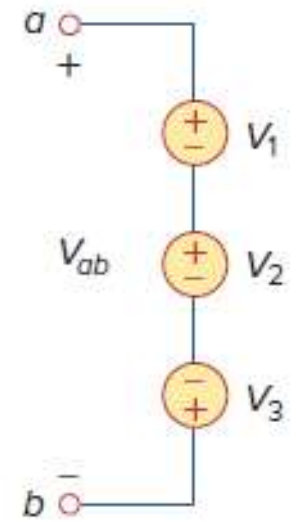
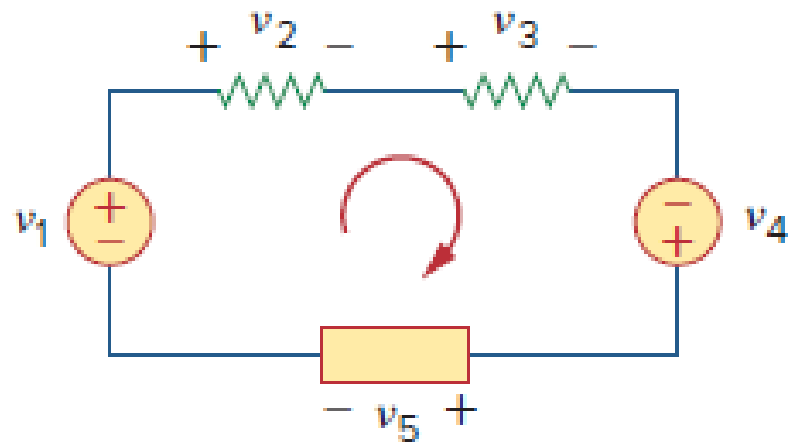
- Find v_1 and v_2 in the circuit shown in Fig. Also calculate i_1 and i_2 and the power dissipated in the $12\text{-}\Omega$ and $40\text{-}\Omega$ resistors.



2. Kirchhoff's Voltage Law(KVL)

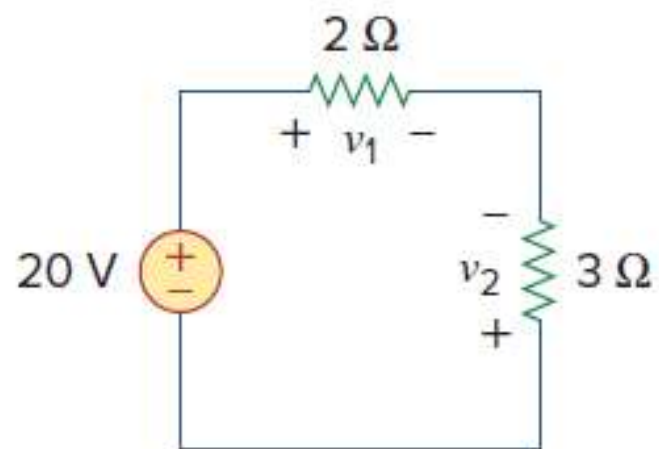
- Kirchhoff's second law is based on the principle of conservation of energy.
- Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.
- Mathematically, KVL implies that $\sum V = 0$, *around a loop*.
- KVL can be applied in two ways: by taking either a clockwise or a counterclockwise trip around the loop. Either way, the algebraic sum of voltages around the loop is zero.
- The sign on each voltage is the polarity of the terminal encountered first as we travel around the loop.



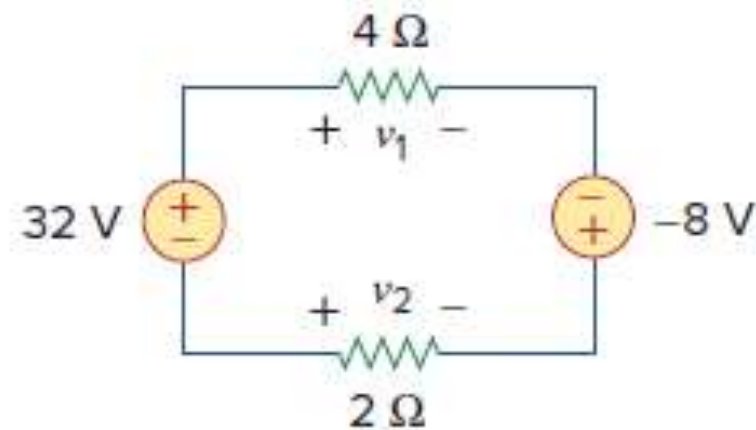


(b)

Problem-1



Problem-2



Series Resistors and Voltage Division

$$V = V_1 + V_2 \quad (KVL)$$

$$V_1 = i \times R_1 ; V_2 = i \times R_2 \quad (Ohms\ Law)$$

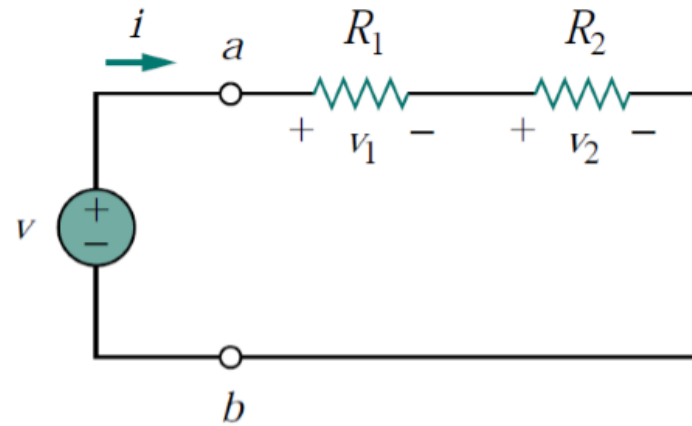
$$V = V_1 + V_2 = (i \times R_1) + (i \times R_2)$$

$$V = V_1 + V_2 = i \times (R_1 + R_2)$$

$$i = \frac{V}{(R_1 + R_2)}$$

$$V_1 = \frac{V}{(R_1 + R_2)} \times R_1$$

$$V_2 = \frac{V}{(R_1 + R_2)} \times R_2$$



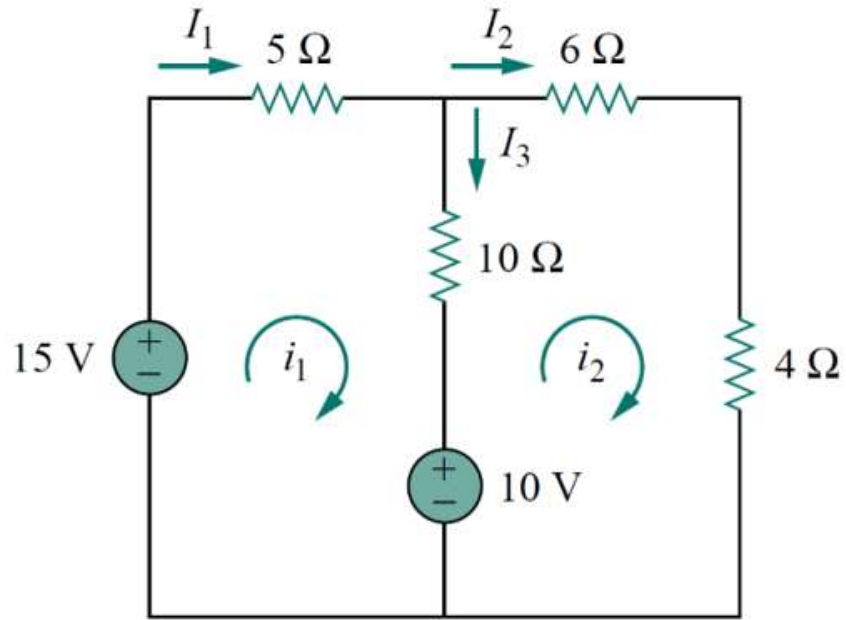
Voltage across a resistor R_x in a series circuit with n resistors

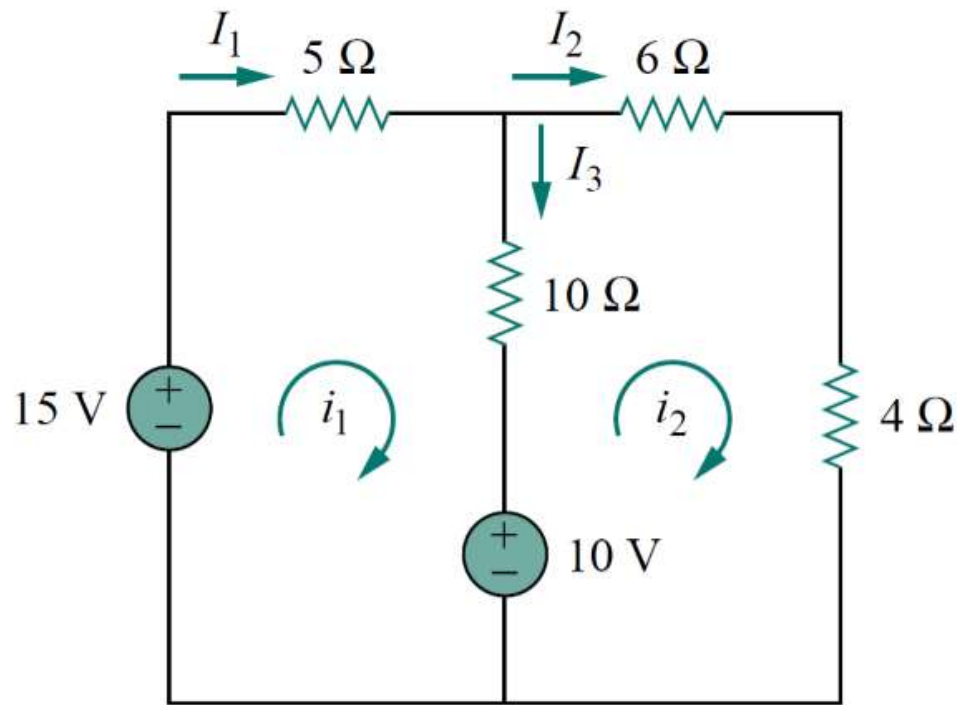
$$V_x = \frac{V}{\sum_{i=1}^n R_i} \times R_x$$

Equivalent resistance of a series circuit with n resistors, R_{Eq}

$$R_{Eq} = \sum_{i=1}^n R_i$$

Problem: find all branch currents





$$15i_1 - 10i_2 = 5 \quad (1)$$

$$-10i_1 + 20i_2 = 10 \quad (2)$$

$$i_1 = 1 \text{ A}; i_2 = 1 \text{ A}$$

$$I_1 = i_1 = 1 \text{ A}; I_2 = i_2 = 1 \text{ A}; I_3 = i_1 - i_2 = 0 \text{ A}$$

$$P_{5\Omega} = I_1^2 \times R = 1^2 \times 5 = 5 \text{ W}; P_{6\Omega} = I_2^2 \times R = 1^2 \times 6 = 6 \text{ W}; P_{10\Omega} = 0 \text{ W};$$

Determine the power dissipated in all the resistors in the circuit using mesh current analysis.

There are two meshes 1 & 2.

The mesh currents are I_1 and I_2 .

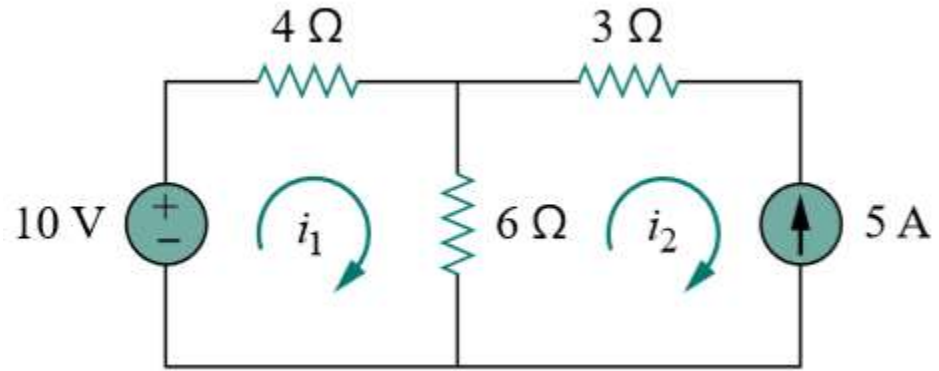
At mesh 1:

$$5i_1 + 10(i_1 - i_2) + 10 - 15 = 0 \quad (1)$$

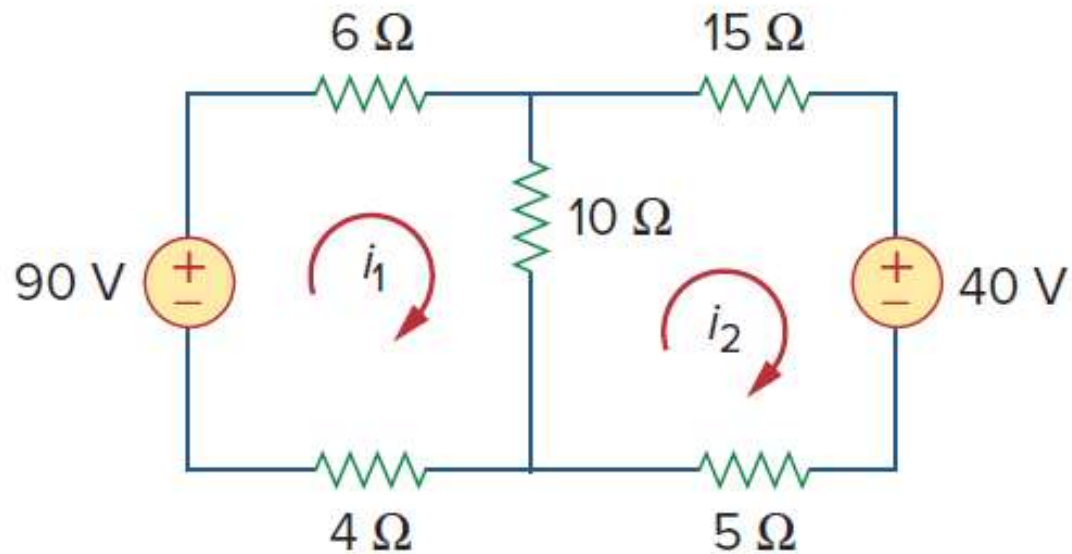
At mesh 2:

$$6i_2 + 4i_2 + -10 - 10(i_1 - i_2) = 0 \quad (2)$$

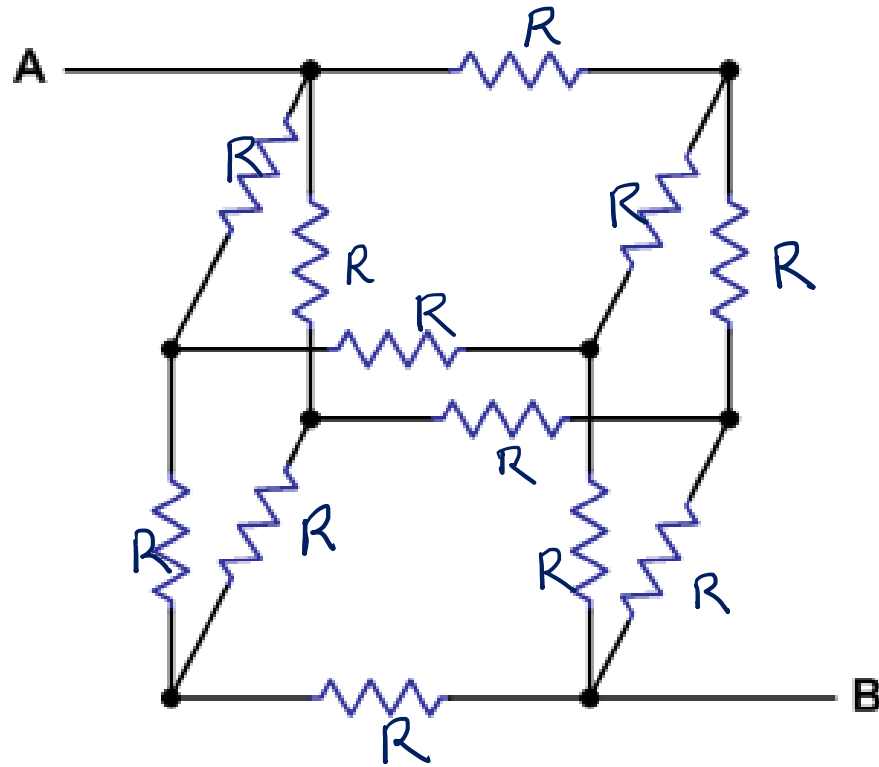
Problem:6: find mesh currents



Calculate the mesh currents i_1 and i_2 of the circuit of Fig

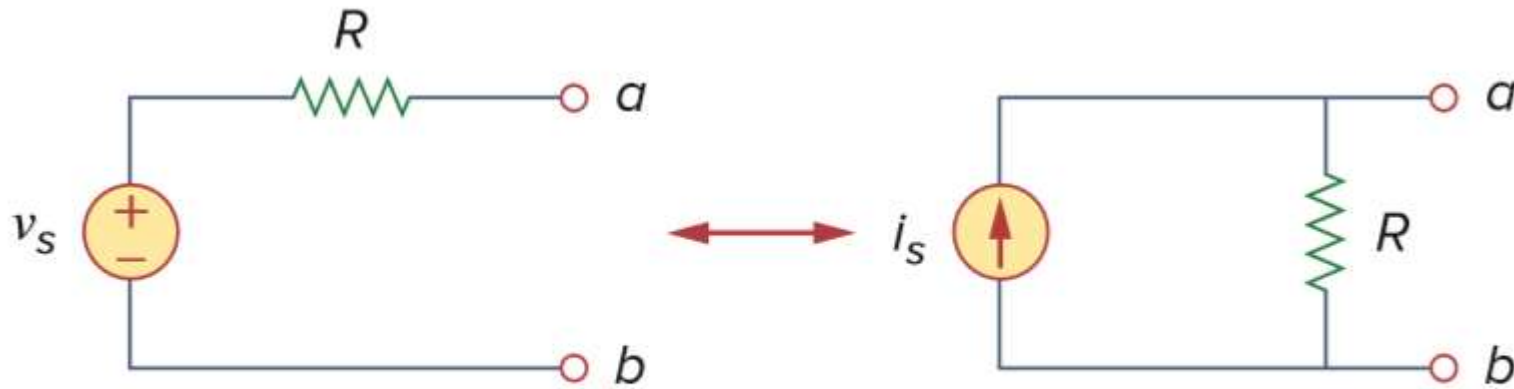


1. Find Equivalent Resistance Across AB terminals



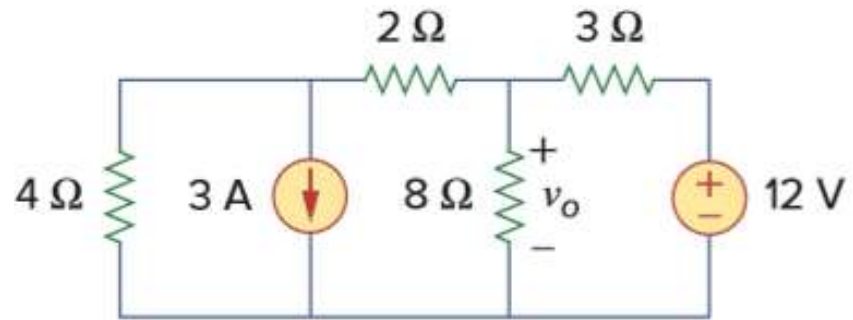
Source Transformation

Source Transformation

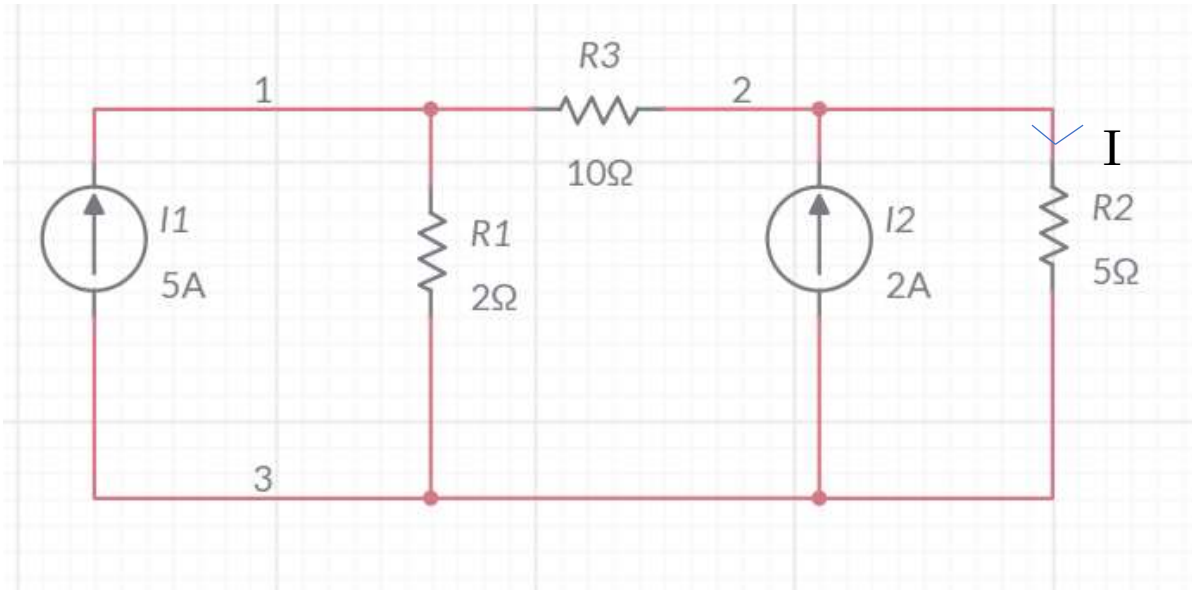


Transformation of independent sources.

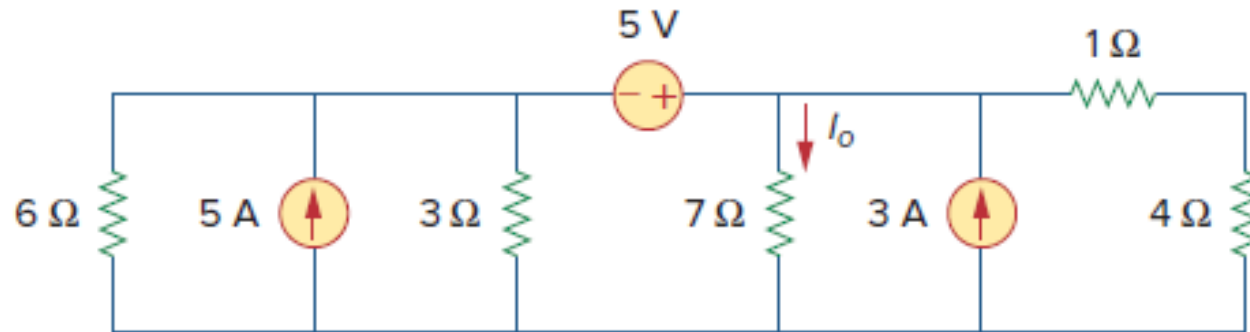
Examples: Find v_o



Problem: find I in the circuit?



Find i_o in the circuit



- Answer: 1.78 A .

Network Theorems

1. Superposition Theorem
2. Thevenin Theorem
3. Maximum Power Transfer Theorem

1. Superposition Theorem

- The superposition principle states that the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone.

